Zhiling Zhou

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Education

Yale University Aug 2025 – Present
PhD Student, Department of Statistics and Data Science

Fudan University Sept 2020 – Jun 2025
B.S. in Data Science and Big Data Technology
School of Data Science, Shanghai, China

Publications and Preprints

1. **How Transformers Represent Hierarchies: A Local-to-Global Mechanism**.
Zhiling Zhou, Tianhao Wang, Zhuoran Yang.
ICML 2026. Paper Link

2. **Demystifying Classifier-Free Guidance for Auto-Regressive Image Generation**.
Zhiling Zhou*, Jiachun Pan*, Fengzhuo Zhang*, Dirk Bergemann, Zhuoran Yang.
Scientific Understanding of Foundation Models Workshop at COLM 2026 (Oral). Paper Link
Mechanistic Interpretability Workshop at ICML 2026. Paper Link

3. **Incremental Gauss–Newton Methods with Superlinear Convergence Rates**.
Zhiling Zhou, Zhuanghua Liu, Chengchang Liu, Luo Luo.
Journal of Optimization Theory and Applications. Paper Link

4. **DiffSim: Aligning Diffusion Model and Molecular Dynamics Simulation for Blind Docking**.
Yanlin Fei, Yurui Li, Zhiling Zhou, Haohan Wang.
IEEE BIBM 2024. Paper Link

5. **Robust Network Learning via Inverse Scale Variational Sparsification**.
Zhiling Zhou*, Zirui Liu*, Chengming Xu, Yanwei Fu, Xinwei Sun.
Preprint. Paper Link

Research Experience

How Transformers Represent Hierarchical Language Structure
Yale University
Advisors: Tianhao Wang, Zhuoran Yang

- Bridge Transformer-based context-free grammar generation with classical parsing algorithms.
- Develop a theoretical framework showing how Transformers implement parsing algorithm.

Demystifying Classifier-Free Guidance for Auto-Regressive Image Generation
Yale University
Advisors: Zhuoran Yang

- Demonstrated the repetition phenomenon and visual degradation in AR models lacking CFG.

- Theoretically modeled how image inductive bias shapes training dynamics and feature representation.
- Submitted to NeurIPS 2026. Preprint version to appear on arXiv.

Incremental Gauss–Newton Method for Nonlinear Equations
Fudan University
Advisor: Luo Luo

- Proposed an incremental Gauss–Newton method with superlinear convergence rates.
- Extended the method with mini-batch updates to improve convergence speed.

Robust Network Learning via Image Sparsification
Fudan University
Advisors: Xinwei Sun, Yanwei Fu

- Developed algorithms based on total variation and efficient Lasso optimization.
- Proposed a graph-based numerical acceleration technique.
- Improved robustness and interpretability of visual learning systems.

Aligning Diffusion Models and Simulations for Blind Docking
University of Illinois Urbana-Champaign
Advisor: Haohan Wang

- Aligned diffusion models with molecular dynamics simulations for blind docking.
- Constructed reliable blind docking datasets using molecular simulations.

Awards

- Third Prize of Excellent Student Scholarship, Fudan University (2020, 2022)
- KLA Scholarship, Fudan University & KLA Corporation (Top 4 undergraduates per year), 2024
- Outstanding Undergraduate Thesis Award, Fudan University, 2025.